

Matrix Group

Notation : In $\mathbb{C}^n$, $\bar{e}_i = (0, \dots, \underset{i^{\text{th}}}{1}, 0 \dots)$
is a vector space basis element.

* An endomorphism α of $\mathbb{C}^n$ is defined as

$$\alpha \bar{e}_i = \sum_{j=1}^n a_{ji} \bar{e}_j$$

where $\alpha = (a_{ij}) \in M_n(\mathbb{C})$.

(We can think of $\alpha \bar{e}_i$ as the i^{th} column vector.)

We use α for a matrix as well as for the endomorphism.

We define multiplication of two endomorphism α, β ^($\alpha \circ \beta$) with corr matrices $(a_{ij}), (b_{ij})$

and construct (c_{ij}) where

$$c_{ij} = \sum_{k=1}^n a_{ik} b_{kj}$$

lemma

$M_n(\mathbb{C}) \cong \mathbb{C}^{n^2}$ as vector spaces.

Proof:

Construct the function

$$\begin{aligned} f: M_n(\mathbb{C}) &\longrightarrow \mathbb{C}^{n^2} \\ (a_{ij}) &\longmapsto (b_{ij}) \end{aligned}$$

where $b_{i+(j-1)n} = a_{ij}$

Observe as j goes from 1 to n , $(j-1)n$ goes from 0 to $(n-1)n$ and as i goes from 1 to n as well, we get $i+(j-1)n$ go from 1 to n^2 .

Observe f is surjective by construction and if $a_{ij} \neq a_{i'j'}$ then $b_{i+(j-1)n} \neq b_{i'+(j'-1)n}$ hence f is bijective.

Observe f is linear hence $M_n(\mathbb{C}) \cong \mathbb{C}^{n^2}$.

□

Remark

Observe we can equip $M_n(\mathbb{C})$ with a topology by ensuring the given map above to be a homeomorphism. By constructing

$$\tau_{M_n(\mathbb{C})} = \{f^{-1}(U) \mid U \in \tau_{\mathbb{C}^{n^2}}\}$$

lemma

Let T be any topological space,

$$\phi: T \longrightarrow M_n(\mathbb{C})$$

ϕ is continuous iff $\pi_{ij} \circ \phi$ is continuous

Proof: Only if condition is trivial.

If condition:

$\pi_{ij} \circ \phi: T \longrightarrow \mathbb{C}$ is continuous.

Now $\prod_{1 \leq i, j \leq n} \pi_{ij} \circ \phi: T \longrightarrow \mathbb{C}^{n^2}$ is

continuous by product topology (or box topology as $\mathbb{C}^{n^2}$ is finite cartesian product).

as $\pi_{ij}^{-1}(U)$ for U open in $\mathbb{C}$ forms a subbasis for product topology on $\mathbb{C}^{n^2}$.

□

Corollary

Product of two matrices is continuous

$$\begin{array}{ccc} \text{Pf: } M_n(\mathbb{C}) \times M_n(\mathbb{C}) & \xrightarrow{q} & M_n(\mathbb{C}) \\ \downarrow f \times f & & \downarrow f \\ \mathbb{C}^{n^2} \times \mathbb{C}^{n^2} & \xrightarrow{p} & \mathbb{C}^{n^2} \end{array}$$

$$\text{where } \begin{array}{ccc} q: M_n(\mathbb{C}) \times M_n(\mathbb{C}) & \longrightarrow & M_n(\mathbb{C}) \\ \downarrow (\sigma, \tau) & \searrow & \downarrow \sigma\tau \end{array}$$

is matrix multiplication. And p is a polynomial

to $p((b_{ij}), (b'_{ij})) = (c_{ij})$ where

$$c_{ij} = \sum_{k=1}^n b_{ik} b'_{kj}. \text{ Hence } p \text{ is continuous.}$$

Therefore $f \circ q = p \circ f \times f$ is continuous.

For any open set in $M_n(\mathbb{C})$ U , we

express $U = f^{-1}(O)$ for some open set O in

$\mathbb{C}^{n^2}$. Hence $q^{-1}(U) = q^{-1}(f^{-1}(0)) = (f \circ q)^{-1}(0)$
 which is open as $f \circ q$ is continuous.

□

Notation: ${}^t\alpha$ is transpose of a matrix and $\overline{\alpha}$ is complex conjugate of a matrix.

lemma

Transpose and complex conjugate are homeomorphism.

Pf:

$$\text{If } h: M_n(\mathbb{C}) \xrightarrow{\alpha} M_n(\mathbb{C}) \text{ ,}$$

$$\alpha \longmapsto {}^t\alpha$$

we can construct $r: \mathbb{C}^{n^2} \longrightarrow \mathbb{C}^{n^2}$ which

is a rearrangement of indices such that

$$b_{i+(j-1)n} = b_{j+(i-1)n} \quad \text{which a homeomorphism.}$$

Hence, $h = f^{-1} \circ r \circ f$ is a homeomorphism.

$$\text{If } k: M_n(\mathbb{C}) \xrightarrow{\alpha} M_n(\mathbb{C}) \text{ is homeomorphic}$$

$$\alpha \longmapsto \overline{\alpha}$$

as k_{ij} is homeomorphic for $1 \leq i, j \leq n$.

Property of matrices

$$(\alpha \beta)^t = \beta^t \alpha^t$$

$$\overline{\alpha \beta} = \overline{\alpha} \cdot \overline{\beta}$$

* A regular / non singular $n \times n$ matrix, if it has an inverse i.e. if there exists a matrix σ^{-1} such that $\sigma \sigma^{-1} = \sigma^{-1} \sigma = \varepsilon$, where ε is the unit matrix of degree n .

A necessary and sufficient condition for σ to be regular is $\det \sigma \neq 0$.

Remarks

If σ is a regular matrix

(i) iff σ^{-1} i.e. reciprocal endomorphism exists

(ii) then ${}^t(\sigma^{-1}) = ({}^t\sigma)^{-1}$, $(\overline{\sigma})^{-1} = \overline{(\sigma^{-1})}$

Remark

If σ, τ are regular matrices then $\sigma\tau$ are regular matrices.

* $\det: M_n(\mathbb{C}) \longrightarrow \mathbb{C}$ can be expressed as

$$\begin{array}{ccc} M_n(\mathbb{C}) & \xrightarrow{\det} & \mathbb{C} \\ f \downarrow & & \nearrow p \\ \mathbb{C}^{n^2} & & \end{array}$$

p is some polynomial function of n^2 variables, which is continuous. We know f is continuous.

Hence, $\det$ map is continuous.

Remark

$$\begin{aligned} GL_n(\mathbb{C}) &= \{ \sigma \in M_n(\mathbb{C}) \mid \det \sigma \neq 0 \} \\ &= M_n(\mathbb{C}) \setminus \det^{-1} \{0\} \end{aligned}$$

is a group and an open subset as $\det$ is a continuous map.

(ii) For $\sigma = (a_{ij}) \in GL_n(\mathbb{C})$ the coefficients b_{ij} of σ^{-1} are given by $b_{ij} = \frac{1}{\det \sigma} A_{ij}$

where A_{ij} is a polynomial of $f\sigma$. Hence mapping $\sigma \mapsto \sigma^{-1}$ is continuous on $GL(n, \mathbb{C})$. Moreover, it's a homeomorphism.

Remark

If $\sigma \in GL_n(\mathbb{C})$, we define $\sigma^* = (\text{tr} \sigma)^{-1}$.

Then we have $(\sigma \tau)^* = \sigma^* \tau^*$. Hence

$\sigma \mapsto \sigma^*$ is a homeomorphism, and an automorphism of order 2 of $GL_n(\mathbb{C})$.

Definitions

$$① O_n(\mathbb{C}) = \{ \sigma \in GL_n(\mathbb{C}) \mid \sigma^* = \sigma \}$$

is set of orthogonal matrices. If $\sigma = \sigma^*$ then σ is complex orthogonal and if $\sigma = \sigma^* = \overline{\sigma}$ then real orthogonal.

$$② U_n(\mathbb{C}) = \{ \sigma \in GL_n(\mathbb{C}) \mid \sigma^* = \overline{\sigma} \}$$

is set of all unitary matrices.

Note $U_n(\mathbb{C})$ or $O_n(\mathbb{C})$ are non empty as $I \in U_n(\mathbb{C})$ and $I \in O_n(\mathbb{C})$.

Notation: $U_n(\mathbb{C}) = U(n)$ and $O_n(\mathbb{R}) = O(n)$.

Don't be confused !

lemma

If f, g are continuous mapping from topological space X to hausdorff space Y then

$\{x \in X \mid f(x) = g(x)\}$ is closed.

Proof:

Consider $\{x \mid f(x) \neq g(x)\}$. Now

$x \in \{x \in X \mid f(x) \neq g(x)\}$ implies $f(x) \neq g(x)$

in Y . By Hausdorff property there exists open

set U_1 and U_2 such that $f(x) \in U_1$ and

$g(x) \in U_2$ and $U_1 \cap U_2 = \emptyset$.

Now $f^{-1}(U_1)$ and $g^{-1}(U_2)$ are open.

Let $V = f^{-1}(U_1) \cap g^{-1}(U_2)$ be a nbd of x .

For any $y \in V$ $f(y) \neq g(y)$ otherwise

$U_1 \cap U_2 \neq \emptyset$. Hence $x \in V \subseteq \{x \mid f(x) \neq g(x)\}$

is open. Hence $\{x \mid f(x) = g(x)\}$ is closed. $\square$

Proposition

$O_n(\mathbb{C})$, $U(n)$, $O(n)$ are closed subgroups of $GL_n(\mathbb{C})$.

Proof.

Observe mapping $\sigma \mapsto \sigma^*$, $\sigma \mapsto \overline{\sigma}$ and identity mapping are continuous.

By the above lemma, as $\mathbb{R}$ is T_2 space.

The sets $U(n) = \{\sigma \mid \overline{\sigma} = \sigma^*\}$ and $O_n(\mathbb{C}) = \{\sigma \mid \sigma = \sigma^*\}$

are closed sets. Observe $GL_n(\mathbb{R}) = \{\sigma \mid \sigma = \overline{\sigma}\}$ is also closed. Hence $U(n)$ and

$O(n) = O_n(\mathbb{C}) \cap GL_n(\mathbb{R})$ is closed.

As all the given sets are non-empty by subgroup test, we can show these are subgroups.

* $SL_n(\mathbb{C}) = \{ \sigma \in GL_n(\mathbb{C}) \mid \det \sigma = 1 \}$
is special linear group.

As $E \in SL_n(\mathbb{C})$, and by subgroup tests
we can show $SL_n(\mathbb{C})$ is a subgroup of
 $GL_n(\mathbb{C})$. Moreover, $SL_n(\mathbb{C}) = \det^{-1}\{1\}$
implies $SL_n(\mathbb{C})$ is a closed subgroup.

Remark:

$$SL(n) = GL_n(\mathbb{R}) \cap SL_n(\mathbb{C})$$

$$SU(n) = SL_n(\mathbb{C}) \cap U(n)$$

$$SO(n) = SL_n(\mathbb{C}) \cap O(n)$$

are closed subgroups of $GL_n(\mathbb{C})$.

Theorem

$U(n)$, $O(n)$, $SU(n)$ and $SO(n)$ are compact.

Proof: Observe $O(n)$, $SU(n)$ and $SO(n)$ are closed subgroups of $U(n)$. It is suffice to show $U(n)$ is compact.

Goal: $U(n)$ is homeomorphic to a closed and bounded set of $\mathbb{C}^{n^2}$.

Observe $GL_n(\mathbb{C})$ is a open subgroup of $M_n(\mathbb{C})$ hence it is also a closed subgroup according to a previous lemma. Observe $U(n)$ is closed in $GL_n(\mathbb{C})$ and $GL_n(\mathbb{C})$ is closed in $M_n(\mathbb{C})$ implies $U(n)$ is closed in $GL_n(\mathbb{C})$.

Now, for $\sigma = (a_{ij}) \in U(n)$ if and only if

$${}^t \sigma \overline{\sigma} = \varepsilon \iff \sum_k a_{ki} \overline{a_{kj}} = \delta_{ij}$$

implies $\sum_{k=1}^n a_{ki} \overline{a_{ki}} = 1$ which means for i^{th} column vector, the absolute sum is 1

this implies for any $1 \leq i, j \leq n$ $|a_{ji}| \leq 1$.

Now in $(\mathbb{C}^{n^2}, \|\cdot\|)$, $v = (v_k)_{k=1}^{n^2}$, $|v_k| \leq 1$

for $1 \leq k \leq n^2$ we define $\|v\| = \max_{1 \leq k \leq n^2} |v_k|$.

Observe $f: M_n(\mathbb{C}) \longrightarrow \mathbb{C}^{n^2}$ as homeomorphism

$f(U(n))$ is closed and bounded, and

compact. This implies $U(n)$ is compact.

□

Note: The choice of the norm does not matter as all norms in finite dimension vector space are equivalent (Add the proof to that in the appendix).